YouTube

ANALYSIS OF GLOBAL YOUTUBE STATISTICS 2023



EMMANUEL EMMANUEL CHIAMAKA

Friday 29th march, 2024

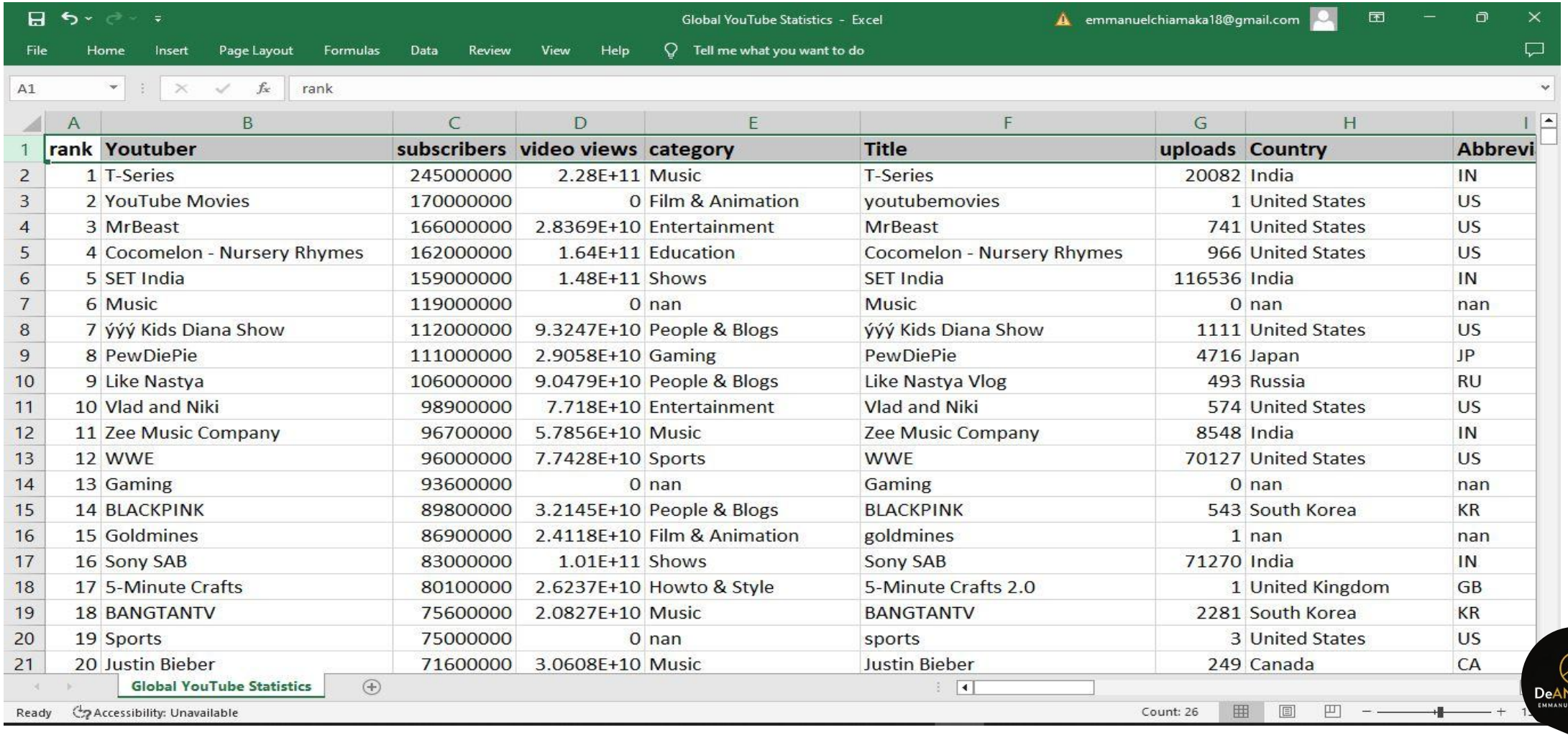
INTRODUCTION

In today's digital age, YouTube stands as a dominant force, shaping how information is shared, entertainment is enjoyed, and communities are formed globally. Understanding YouTube's usage dynamics, trends, and user behavior is crucial for content creators, marketers, researchers, and policymakers. This project focuses on analyzing global YouTube statistics using advanced data analytics to uncover actionable insights. By delving into metrics like viewership patterns, engagement levels, demographics, and content preferences, the aim is to provide comprehensive insights beyond surface-level observations. These insights can empower decision-making for content creators, advertisers, and policymakers, driving innovation and fostering a more inclusive digital environment. Leveraging cutting-edge analytics, this project aims to extract valuable insights to inform strategies and policies in the ever-evolving landscape of online content consumption.

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BASICS

The dataset used in this project was obtained from Kaggle. It was meticulously compiled from various reputable sources, ensuring accuracy and reliability of the information presented. The survey results were collated on a spreadsheet (Microsoft Excel) with column heads which include:



rank	Youtuber	subscribers	video views	category	Title	uploads	Country	Abbrevi
1	T-Series	245000000	2.28E+11	Music	T-Series	20082	India	IN
2	YouTube Movies	170000000	0	Film & Animation	youtubemovies	1	United States	US
3	MrBeast	166000000	2.8369E+10	Entertainment	MrBeast	741	United States	US
4	Cocomelon - Nursery Rhymes	162000000	1.64E+11	Education	Cocomelon - Nursery Rhymes	966	United States	US
5	SET India	159000000	1.48E+11	Shows	SET India	116536	India	IN
6	Music	119000000	0	nan	Music	0	nan	nan
7	ýýý Kids Diana Show	112000000	9.3247E+10	People & Blogs	ýýý Kids Diana Show	1111	United States	US
8	PewDiePie	111000000	2.9058E+10	Gaming	PewDiePie	4716	Japan	JP
9	Like Nastya	106000000	9.0479E+10	People & Blogs	Like Nastya Vlog	493	Russia	RU
10	Vlad and Niki	98900000	7.718E+10	Entertainment	Vlad and Niki	574	United States	US
11	Zee Music Company	96700000	5.7856E+10	Music	Zee Music Company	8548	India	IN
12	WWE	96000000	7.7428E+10	Sports	WWE	70127	United States	US
13	Gaming	93600000	0	nan	Gaming	0	nan	nan
14	BLACKPINK	89800000	3.2145E+10	People & Blogs	BLACKPINK	543	South Korea	KR
15	Goldmines	86900000	2.4118E+10	Film & Animation	goldmines	1	nan	nan
16	Sony SAB	83000000	1.01E+11	Shows	Sony SAB	71270	India	IN
17	5-Minute Crafts	80100000	2.6237E+10	Howto & Style	5-Minute Crafts 2.0	1	United Kingdom	GB
18	BANGTANTV	75600000	2.0827E+10	Music	BANGTANTV	2281	South Korea	KR
19	Sports	75000000	0	nan	sports	3	United States	US
20	Justin Bieber	71600000	3.0608E+10	Music	Justin Bieber	249	Canada	CA

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QUESTIONS CONSIDERED BEFORE THE ANLAYSIS AND VISUALIZATION

1. **YouTube Analytics:**

- What are the success factors of top YouTube channels, and how do they differ from other channels?
- What metrics can we use to measure channel success, and how do they contribute to overall performance?

2. **Content Strategy:**

- Which categories and upload frequencies resonate most with YouTube audiences?
- How do content strategies vary across different categories, and what impact do they have on viewership and engagement?

3. **Regional Influencers:**

- Who are the influential YouTube creators in various countries, and how do they influence global trends?
- What factors contribute to the success of regional influencers, and how does their impact compare on a global scale?

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4. Earnings Analysis:

- Is there a correlation between channel performance metrics (e.g., views, subscribers) and estimated earnings?
- How do different factors, such as content type and audience demographics, affect a channel's earning potential?

5. **Geospatial Visualization:**

- What is the distribution of successful YouTube channels across the world, and are there any geographical trends?
- How does the geographical location of creators impact their success and audience reach?

6. **Trending Topics:**

- How do certain categories gain popularity over time, and what role do world events play in shaping these trends?
- Can we identify patterns in content trends, and how do they evolve over different time periods?

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PROJECT STRATEGY

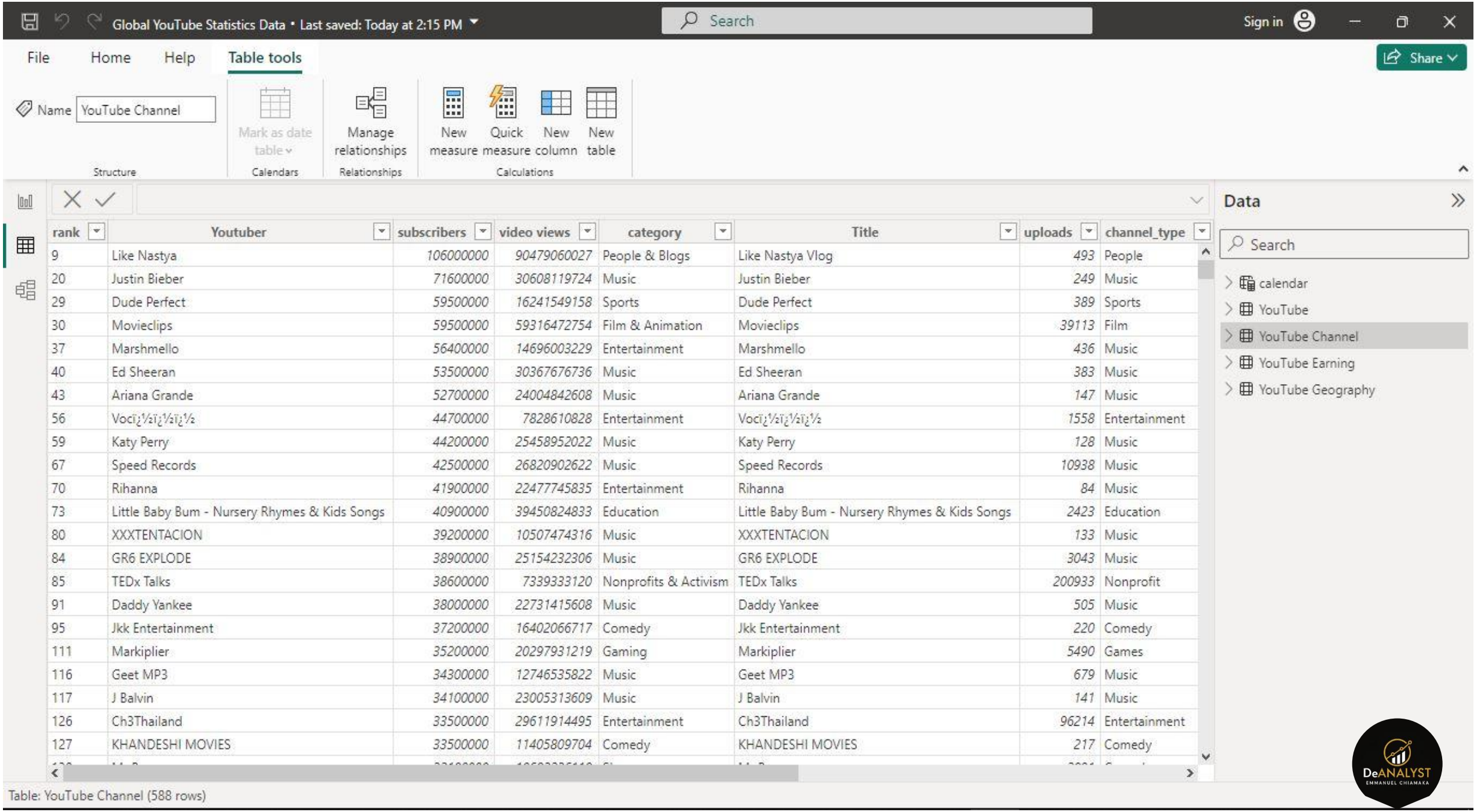
The project commenced with the data upload to Power BI, after some cleaning and grouping in Excel, more rigorous cleaning and preparation followed using Power Query. This ensured data accuracy and consistency, with operations like removing duplicates and standardizing formats being executed.

To streamline analysis, duplicate tables were created with distinct titles such as YOUTUBE, YOUTUBE EARNINGS, YOUTUBE CHANNELS, and YOUTUBE GEOGRAPHY. This organizational structure facilitated a structured exploration of different dataset aspects.

Within the YOUTUBE GEOGRAPHY table, geographical data was enhanced by adding latitude and longitude columns using DAX measures. **While this enabled geospatial visualization and analysis, limitations arose due to the need for a Power BI Services account to access certain features like Map Display Visualization.**

Comprehensive documentation accompanied each step of the process, ensuring transparency and clarity in the analysis. Limitations, particularly those concerning geographical visualization capabilities, were acknowledged to provide a clear roadmap for further analysis and decision-making.

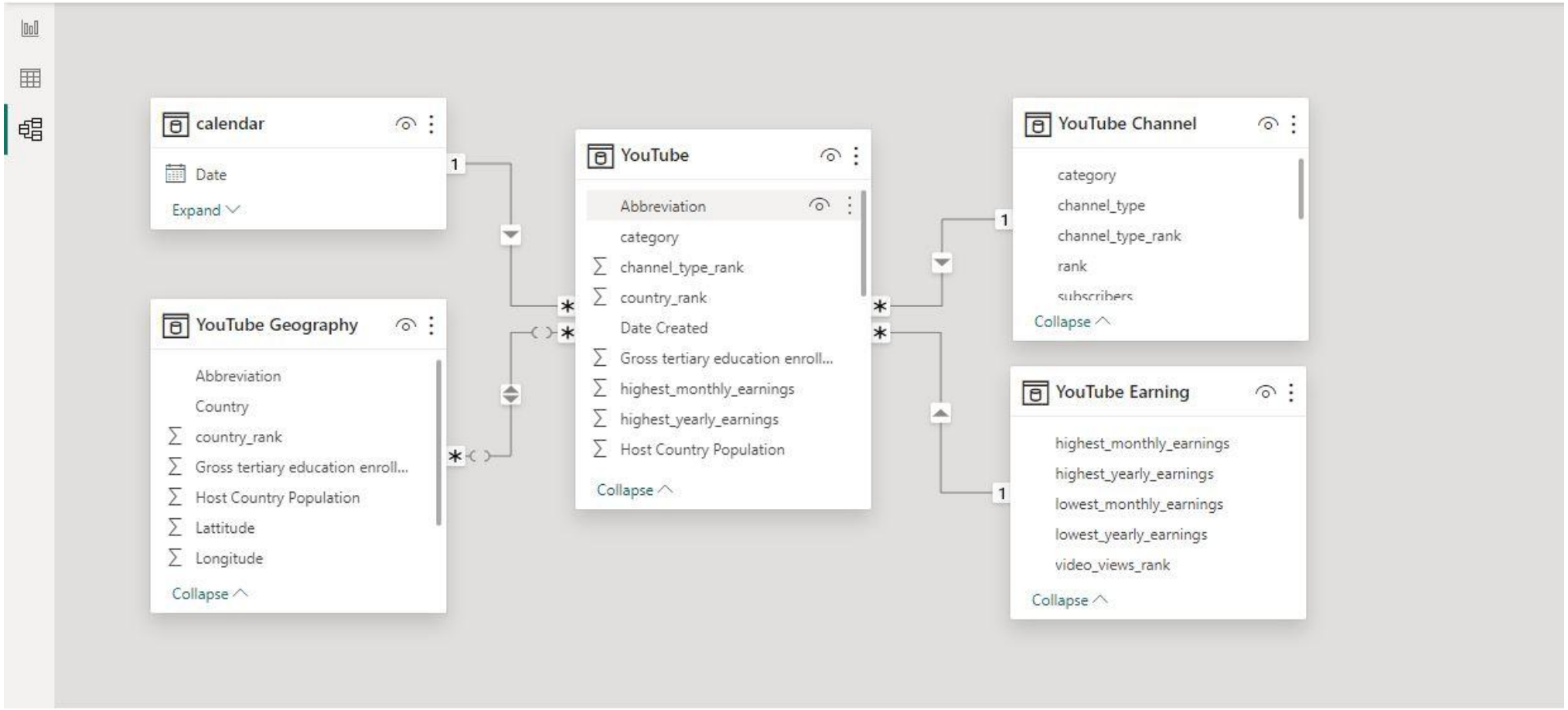
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rank	Youtuber	subscribers	video views	category	Title	uploads	channel_type
9	Like Nastya	106000000	90479060027	People & Blogs	Like Nastya Vlog	493	People
20	Justin Bieber	71600000	30608119724	Music	Justin Bieber	249	Music
29	Dude Perfect	59500000	16241549158	Sports	Dude Perfect	389	Sports
30	Movieclips	59500000	59316472754	Film & Animation	Movieclips	39113	Film
37	Marshmello	56400000	14696003229	Entertainment	Marshmello	436	Music
40	Ed Sheeran	53500000	30367676736	Music	Ed Sheeran	383	Music
43	Ariana Grande	52700000	24004842608	Music	Ariana Grande	147	Music
56	Vocï¿½ï¿½ï¿½	44700000	7828610828	Entertainment	Vocï¿½ï¿½ï¿½	1558	Entertainment
59	Katy Perry	44200000	25458952022	Music	Katy Perry	128	Music
67	Speed Records	42500000	26820902622	Music	Speed Records	10938	Music
70	Rihanna	41900000	22477745835	Entertainment	Rihanna	84	Music
73	Little Baby Bum - Nursery Rhymes & Kids Songs	40900000	39450824833	Education	Little Baby Bum - Nursery Rhymes & Kids Songs	2423	Education
80	XXXTENTACION	39200000	10507474316	Music	XXXTENTACION	133	Music
84	GR6 EXPLODE	38900000	25154232306	Music	GR6 EXPLODE	3043	Music
85	TEDx Talks	38600000	7339333120	Nonprofits & Activism	TEDx Talks	200933	Nonprofit
91	Daddy Yankee	38000000	22731415608	Music	Daddy Yankee	505	Music
95	Jkk Entertainment	37200000	16402066717	Comedy	Jkk Entertainment	220	Comedy
111	Markiplier	35200000	20297931219	Gaming	Markiplier	5490	Games
116	Geet MP3	34300000	12746535822	Music	Geet MP3	679	Music
117	J Balvin	34100000	23005313609	Music	J Balvin	141	Music
126	Ch3Thailand	33500000	29611914495	Entertainment	Ch3Thailand	96214	Entertainment
127	KHANDESHI MOVIES	33500000	11405809704	Comedy	KHANDESHI MOVIES	217	Comedy

Data Modelling

After obtaining the results above, we closed and applied the results to Power BI, therefore obtaining a model. My model procedure was to create a connection between the YouTube table (fact table) and the YouTube Channel using RANK, YouTube table (fact table) and the YouTube Earning using VIDEO VIEW RANK, YouTube table (fact table) and the YouTube Geography using ABBREVIATION, YouTube table (fact table) and the Calendar using DATE.



DAX MEASURE

In my analysis, DAX measures proved indispensable for executing numerical calculations and ensuring the accuracy of my findings. Leveraging DAX measures was pivotal in maintaining the precision and reliability of my analytical process.

DAX measures were instrumental in various aspects of the analysis, including:

1. Adding two columns (Latitude and Longitude) to the YouTube Geography Table.

2. Calculating the total yearly earnings, considering only the highest and lowest yearly earnings were provided in the dataset.

3. Determining the total monthly earnings, taking into account the highest and lowest monthly earnings provided in the dataset.

4. Utilizing DAX Calendar measures to segment dates into years, quarters, months, and days for comprehensive analysis.

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VISUALIZATION

- Visualizations were created, including **pie charts**, **bar charts, line charts, cards, column charts, tree map etc.**

- This helped in the generation of reasonable insights and providing answers to our questions.

- An interactive dashboard was also formed with **slicers** to ease filtering of the different variables under study

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YouTube HOME

The YouTube HOME visualization serves as the gateway to the analytics project, seamlessly blending the essence of the YouTube brand with intuitive navigation to its various sections. Its primary purpose is to create a visually appealing representation that embodies the identity of YouTube, evoking its familiar atmosphere while facilitating exploration of different analytics segments.





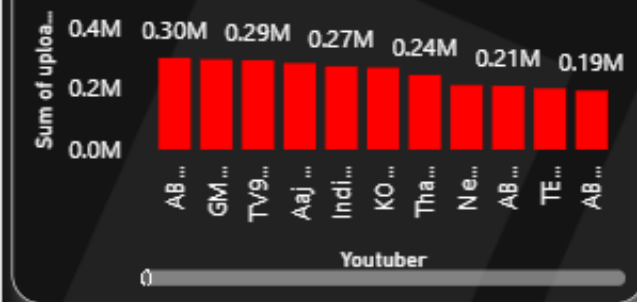
YouTube DASHBOARD

The Dashboard provides valuable insights into key metrics such as Total YouTubers, Categories, uploads, subscribers, and video views, offering a comprehensive overview of YouTube's ecosystem. It distinctly showcases the distribution of YouTubers across different geographical locations, as well as the sum of yearly earnings, uploads, and video views. Additionally, the Dashboard highlights the average number of uploads per YouTuber and the cumulative earnings generated by YouTubers over the analyzed years. This user-friendly interface not only presents essential statistics but also facilitates a deeper understanding of YouTube's global reach and economic impact, making it a valuable tool for stakeholders seeking actionable insights in the realm of online content creation and consumption.

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Total Youtubers	Total Categories	Total Uploads	Total Subscribers	Total Video Views
588	18	8M	14bn	8T

Sum of uploads by Youtuber



Youtuber by Country



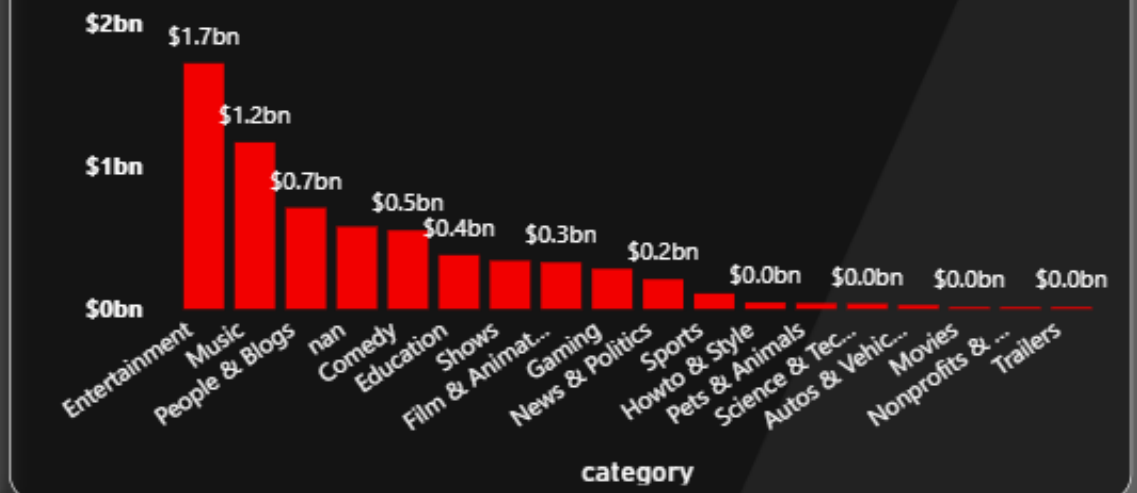
Uploads and video views by category



Sum of Yearly Earnings



Sum Yearly Earnings by Category

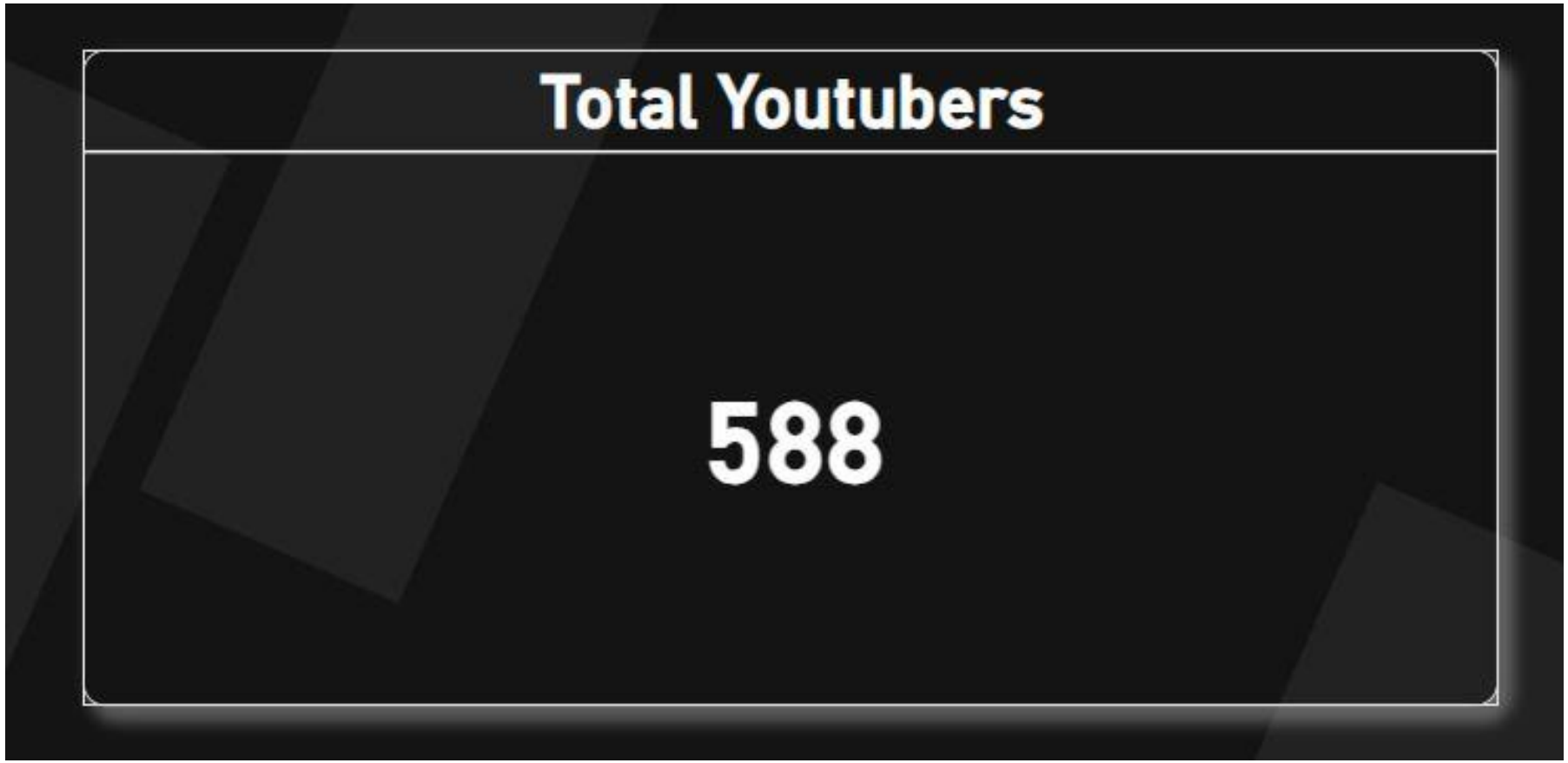


Country

- Argentina
- Australia
- Barbados
- Brazil
- Canada
- Chile
- China
- Colombia
- Cuba
- Ecuador
- Egypt
- El Salvador
- France
- Germany
- India
- Indonesia
- Italy
- Japan
- Jordan
- Kuwait
- Latvia
- Malaysia
- Mexico

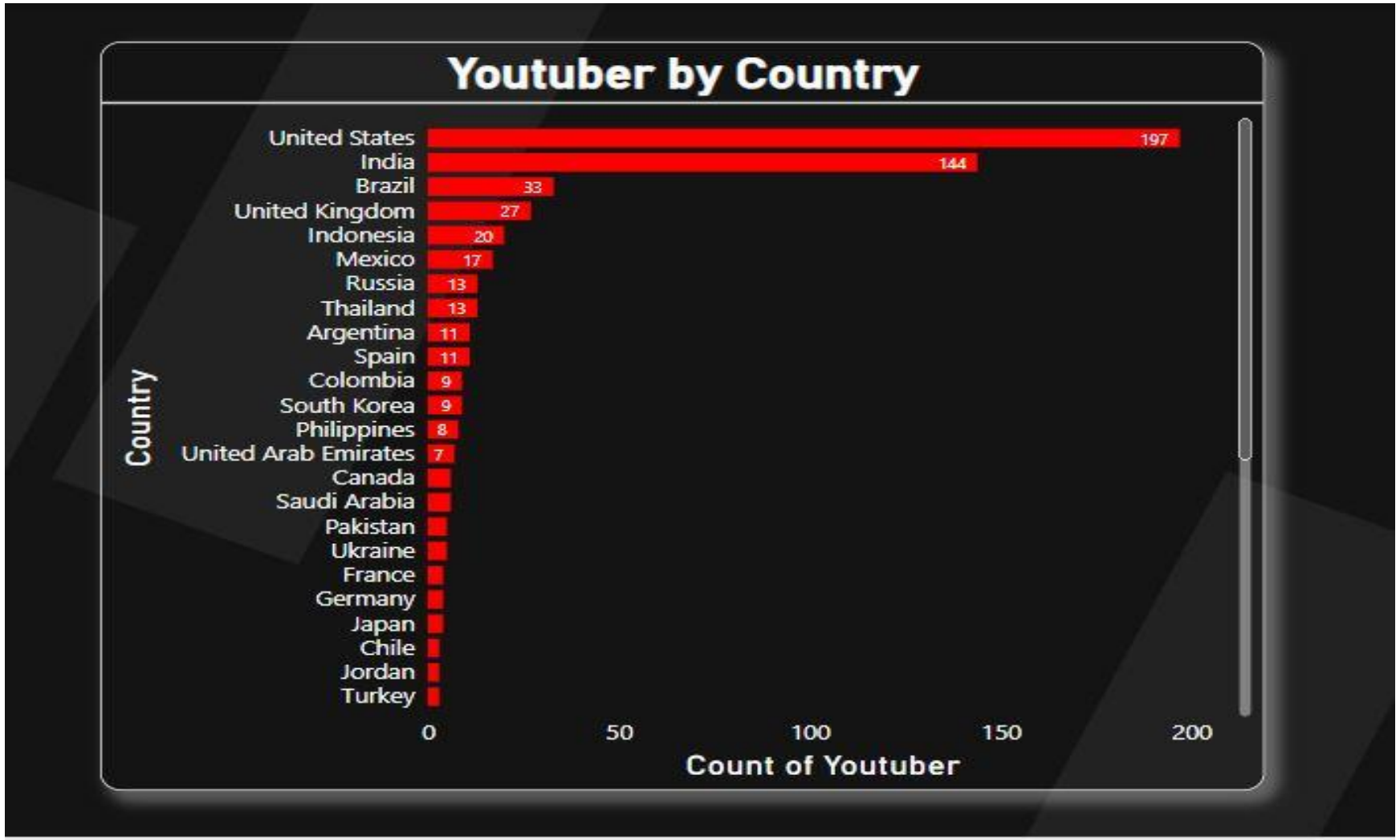
Insight

588 distinct YouTubers are active on the YouTube Platform.



Insight

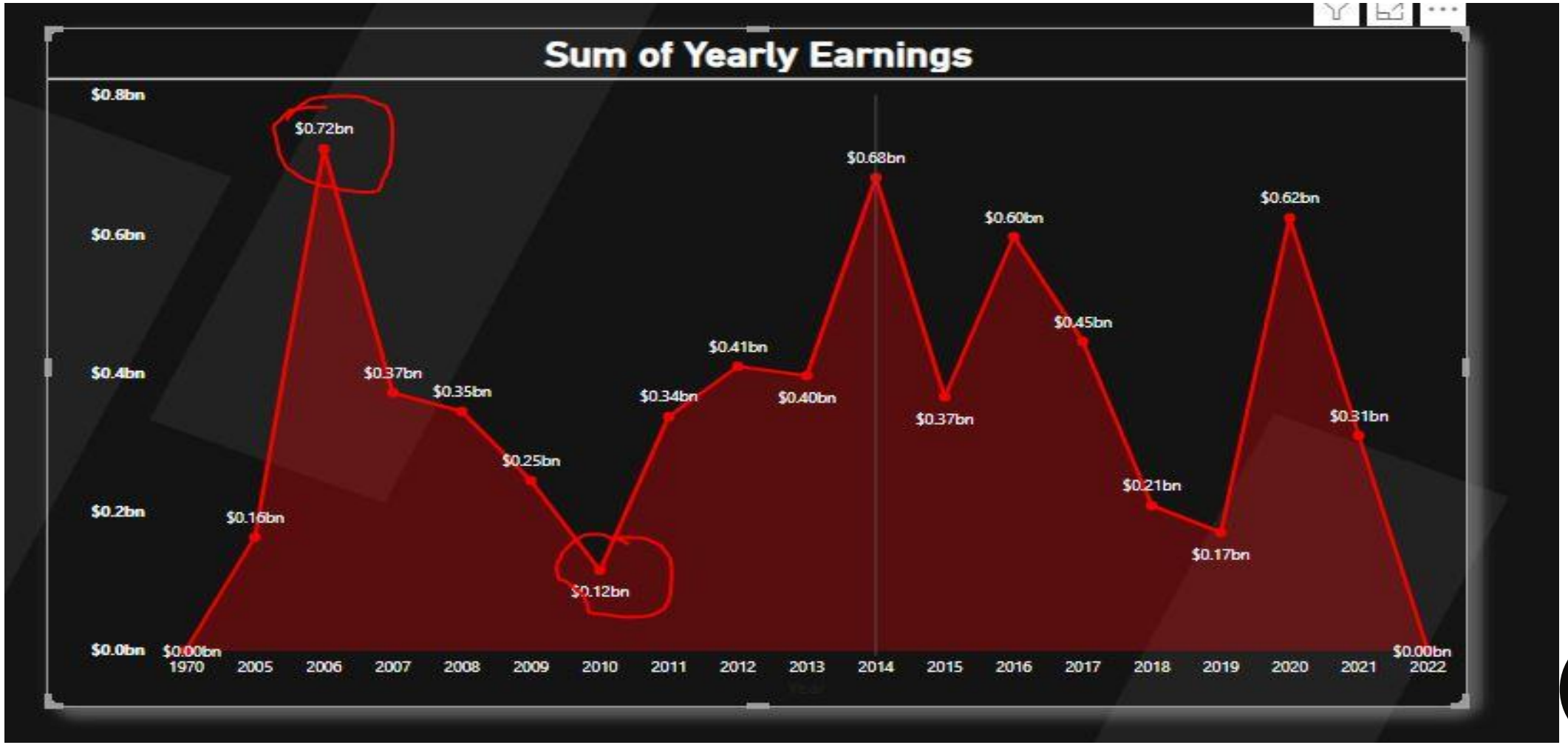
The United States is home to the majority of YouTubers with 197 of the Total 588 Youtubers.



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Insight

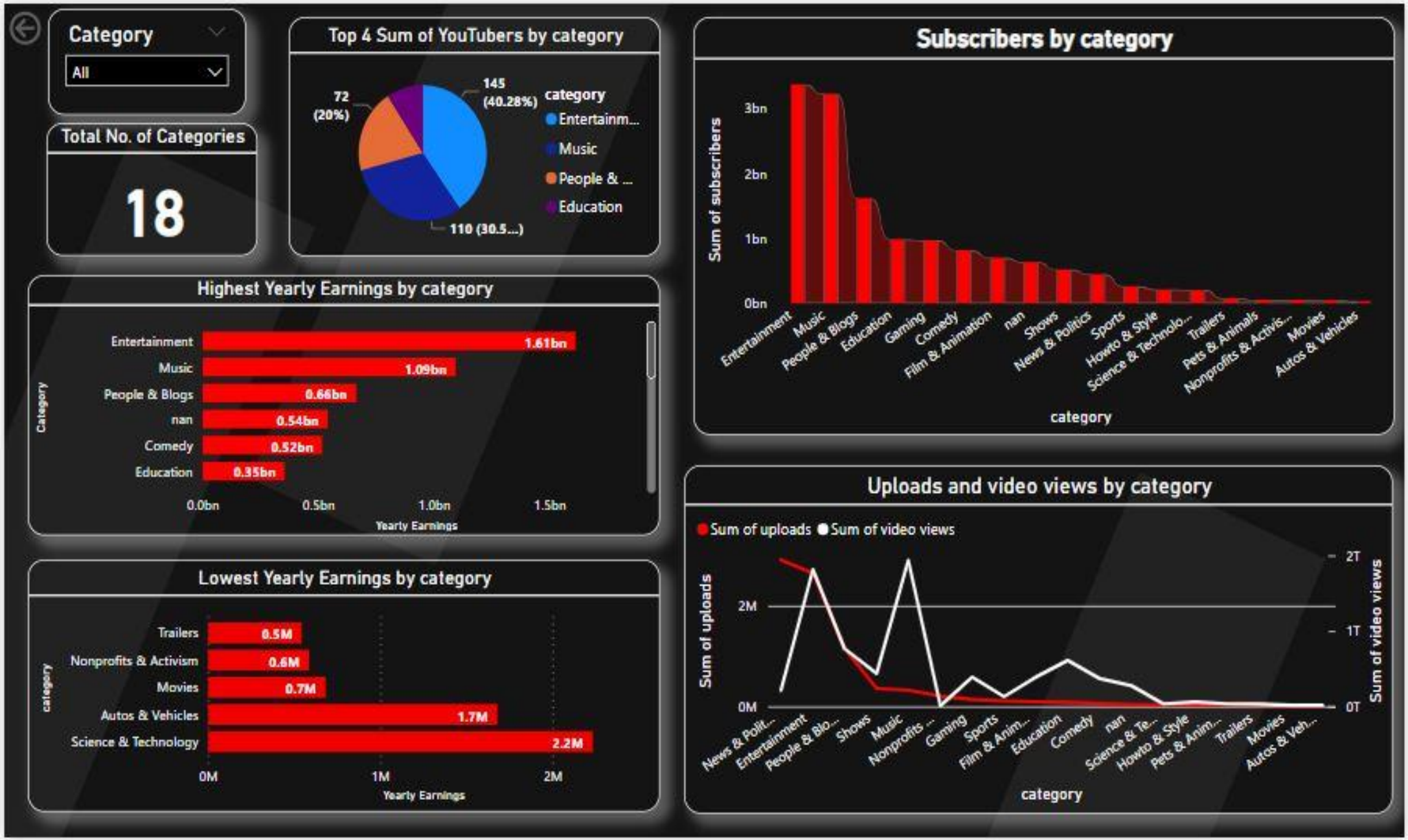
The year 2016 witnessed the highest industry payout to YouTubers, totaling $0.72 billion.



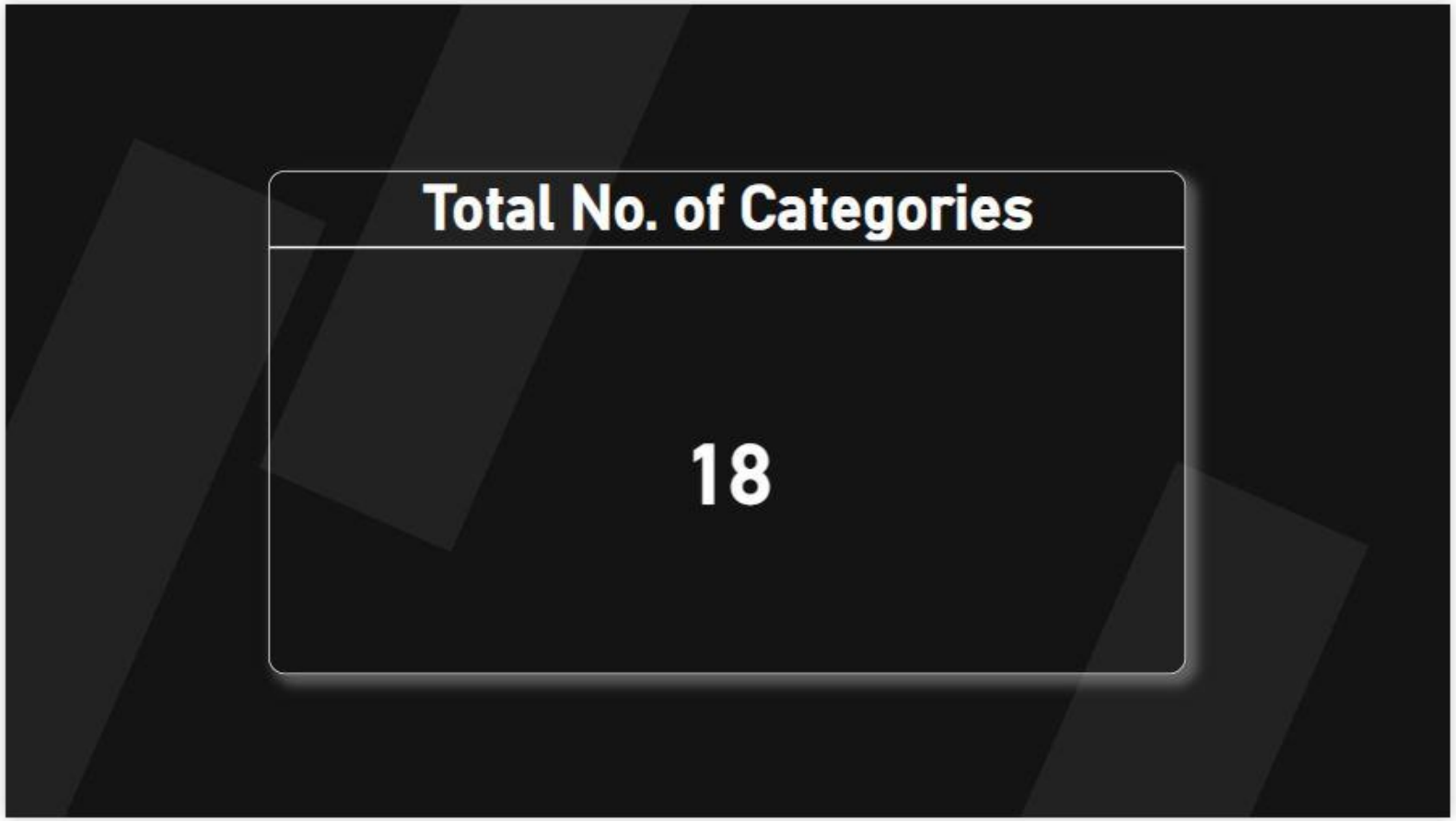
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YouTube CATEGORY ANALYTICS

The YouTube CATEGORY ANALYTICS visualization offers a detailed exploration of various categories, providing answers to key questions such as the total number of categories involved, the distribution of YouTubers across categories, and the metrics associated with each category. It highlights which categories boast the highest number of YouTubers, subscribers, uploads, and views, shedding light on the popularity and engagement levels within each genre. Additionally, the visualization identifies the categories with the highest and lowest yearly earnings, offering valuable insights into the economic landscape of YouTube content creation.

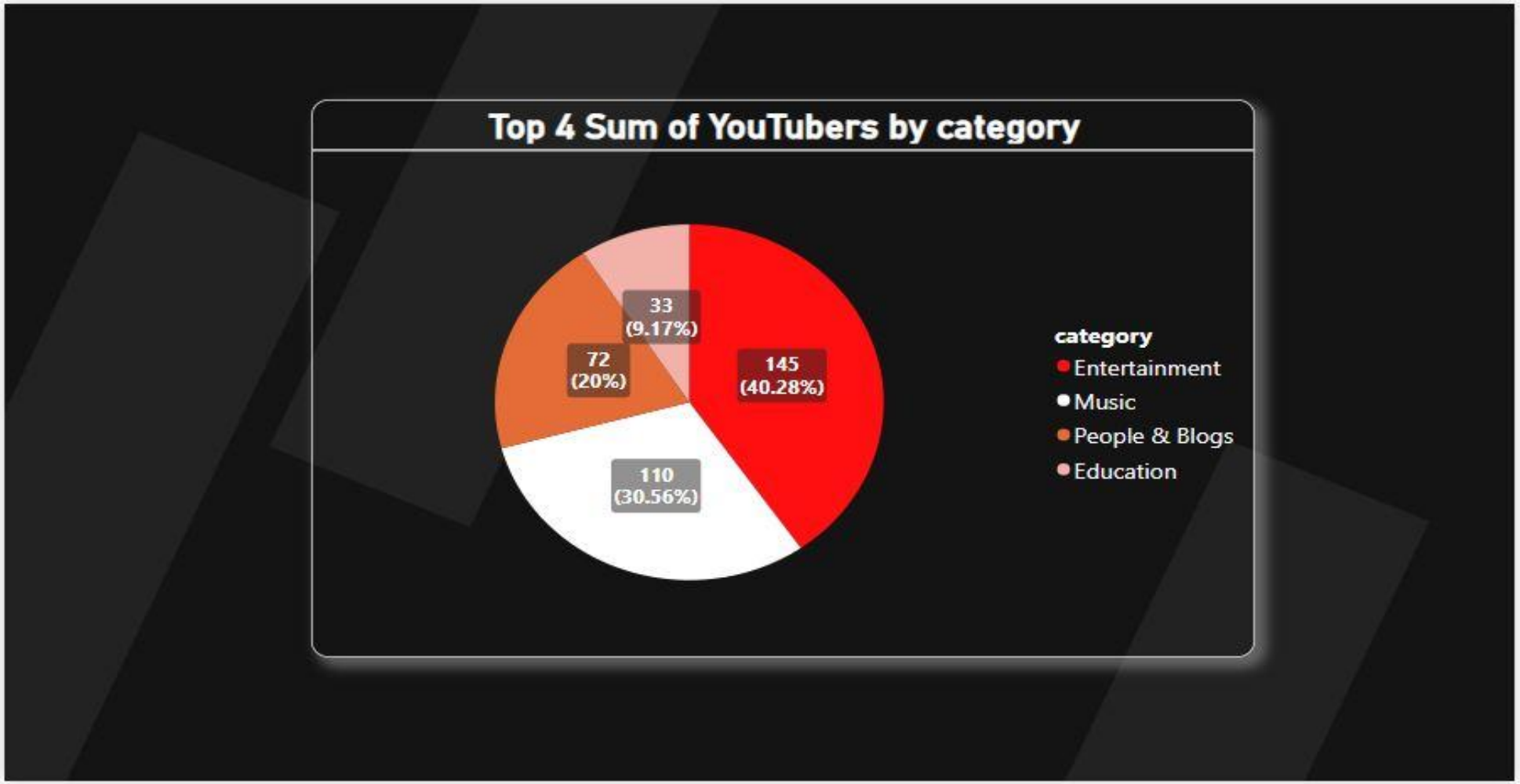

Category
All
Total No. of Categories
18
Top 4 Sum of YouTubers by category
145
(40.28%)
72
(20%)
110 (30.5...)
category
Entertainm...
Music
People & ...
Education
Subscribers by category
Sum of subscribers
3bn
2bn
1bn
0bn
Entertainment
Music
People & Blogs
Education
Gaming
Comedy
Film & Animation
nan
Shows
News & Politics
Sports
Howto & Style
Science & Technolo...
Trailers
Pets & Animals
Nonprofits & Activis...
Movies
Autos & Vehicles
category
Highest Yearly Earnings by category
Category
Entertainment
1.61bn
Music
1.09bn
People & Blogs
0.66bn
nan
0.54bn
Comedy
0.52bn
Education
0.35bn
0.0bn
0.5bn
1.0bn
1.5bn
Yearly Earnings
Lowest Yearly Earnings by category
category
Trailers
0.5M
Nonprofits & Activism
0.6M
Movies
0.7M
Autos & Vehicles
1.7M
Science & Technology
2.2M
0M
1M
2M
Yearly Earnings
Uploads and video views by category
Sum of uploads
Sum of video views
Sum of uploads
2M
0M
2T
1T
0T
Sum of video views
News & Polit...
Entertainment
People & Blo...
Shows
Music
Nonprofits ...
Gaming
Sports
Film & Anim...
Education
Comedy
nan
Science & Te...
Howto & Style
Pets & Anim...
Trailers
Movies
Autos & Veh...
category

Total YouTube Category captured in Data set



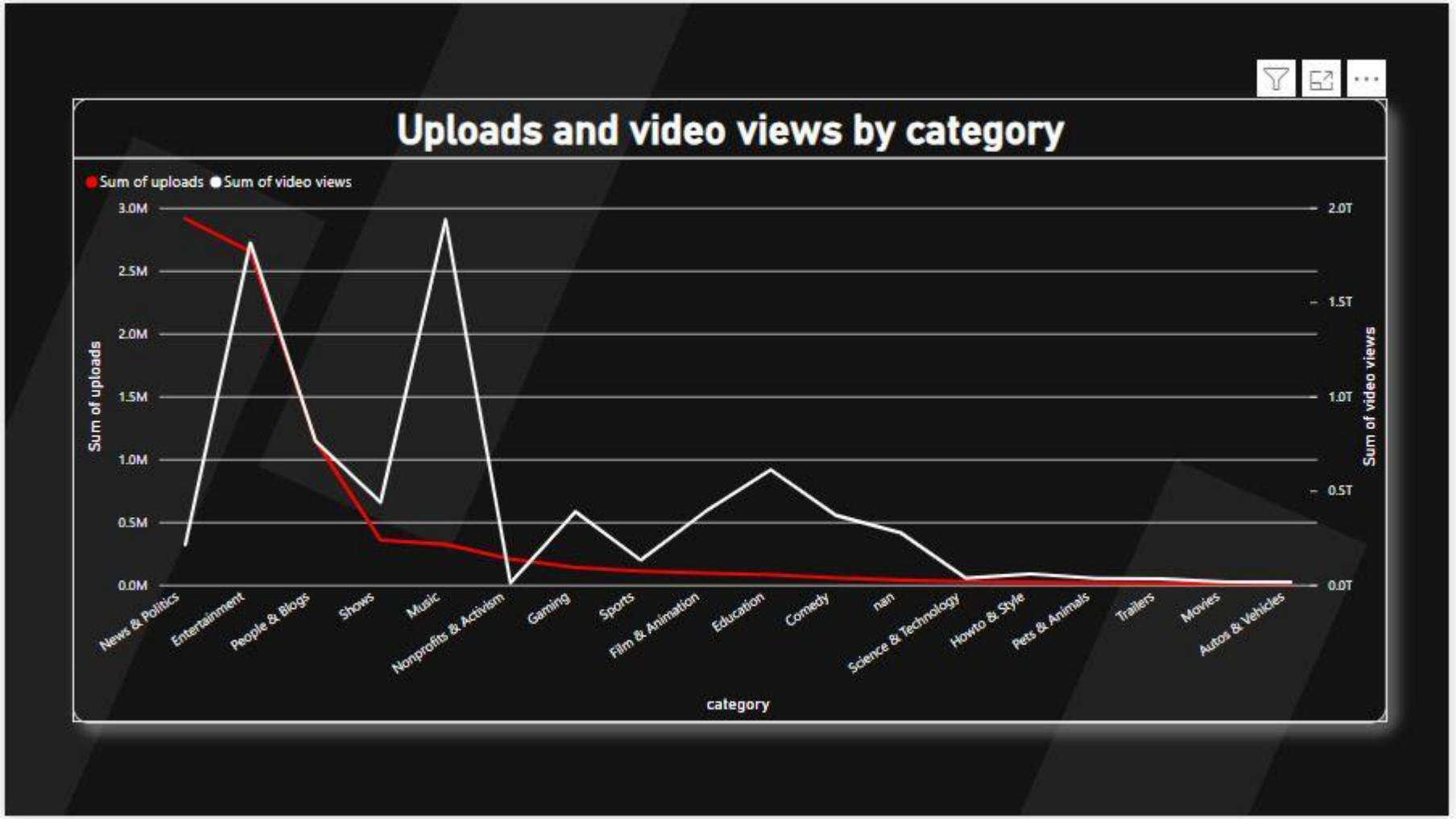
Top 4 Category by number of YouTubers

Among the top four YouTubers by category, Entertainment dominates with 40.28%, followed by Music at 30.56%, People & Blog at 20%, and Education at 9.17%.



Uploads and views by category:

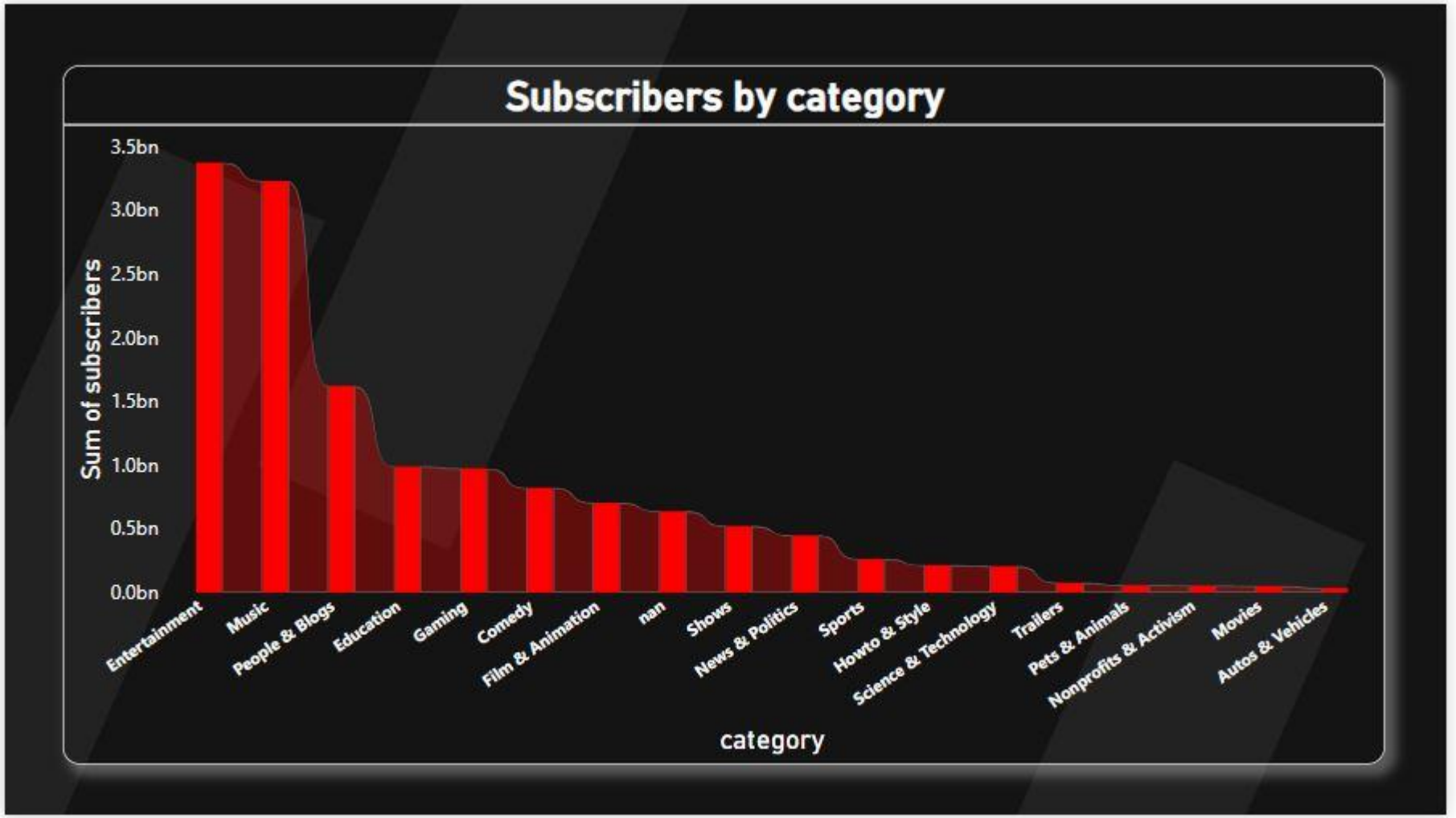
Upload frequency doesn't guarantee views; viewer interest dictates engagement. Music and entertainment, being the most viewed categories, demonstrate strong user preference.



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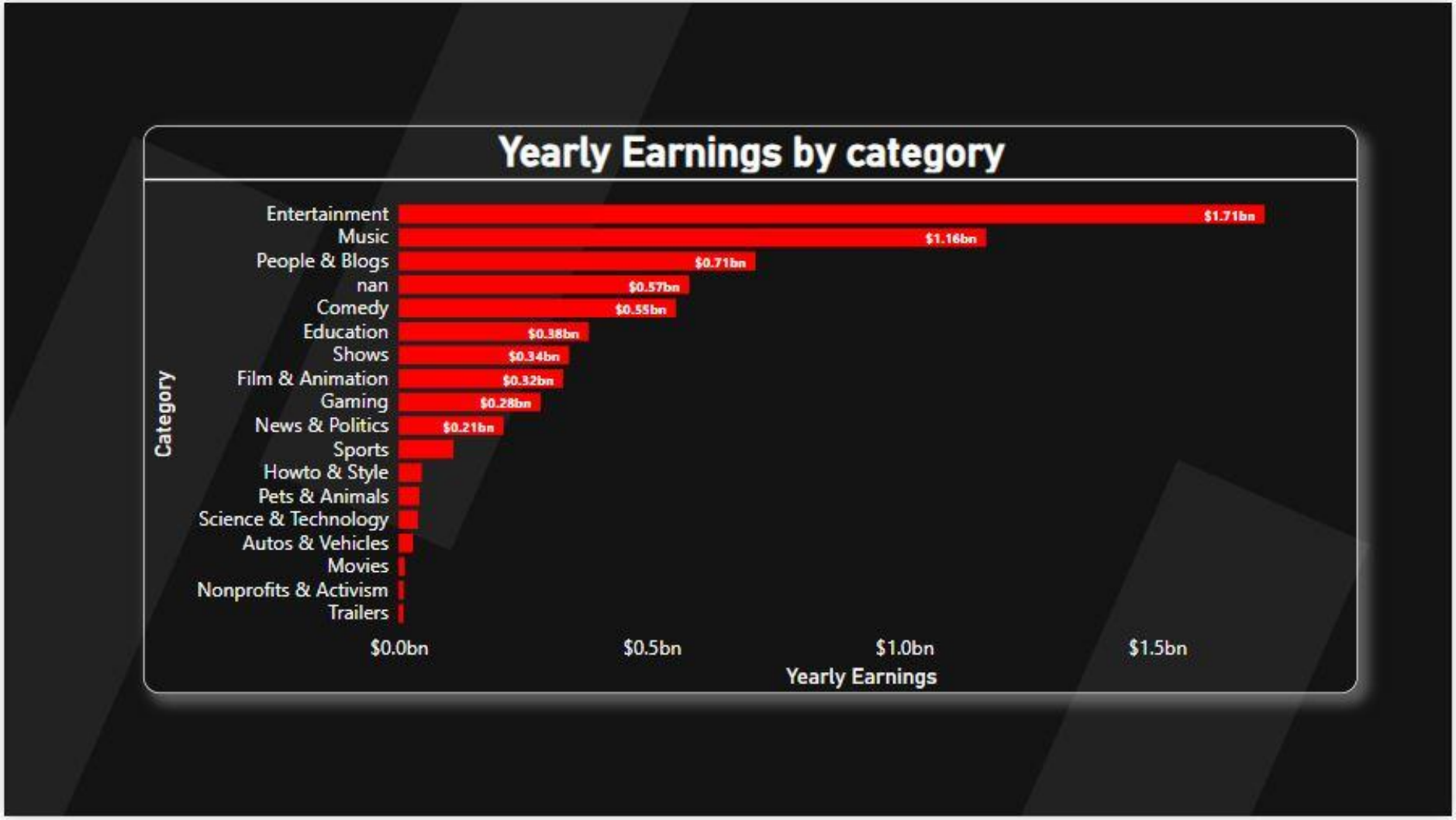
Subscribers by Category

Subscribers play a role in determining category earnings on YouTube; Compare with the yearly earning of categories in the next slide.



Yearly Earnings by Category:

we could confirm that the subscribers has a way of affecting the earning.



YouTube EARNING ANALYTICS

The YouTube EARNING ANALYTICS visualization offers an in-depth exploration of earnings data, addressing key questions surrounding YouTube's financial landscape. It provides insights into total yearly and monthly earnings from the platform, along with a breakdown of yearly earnings by category. Additionally, it delves into YouTuber-specific metrics, offering a glimpse into monthly earnings for individual content creators. The visualization further identifies top earning countries on YouTube based on the provided data. To enhance user experience, a slicer feature is incorporated, enabling quick access to each YouTuber's yearly and monthly earnings, streamlining the process of analyzing individual performance

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EARNING ANALYTICS

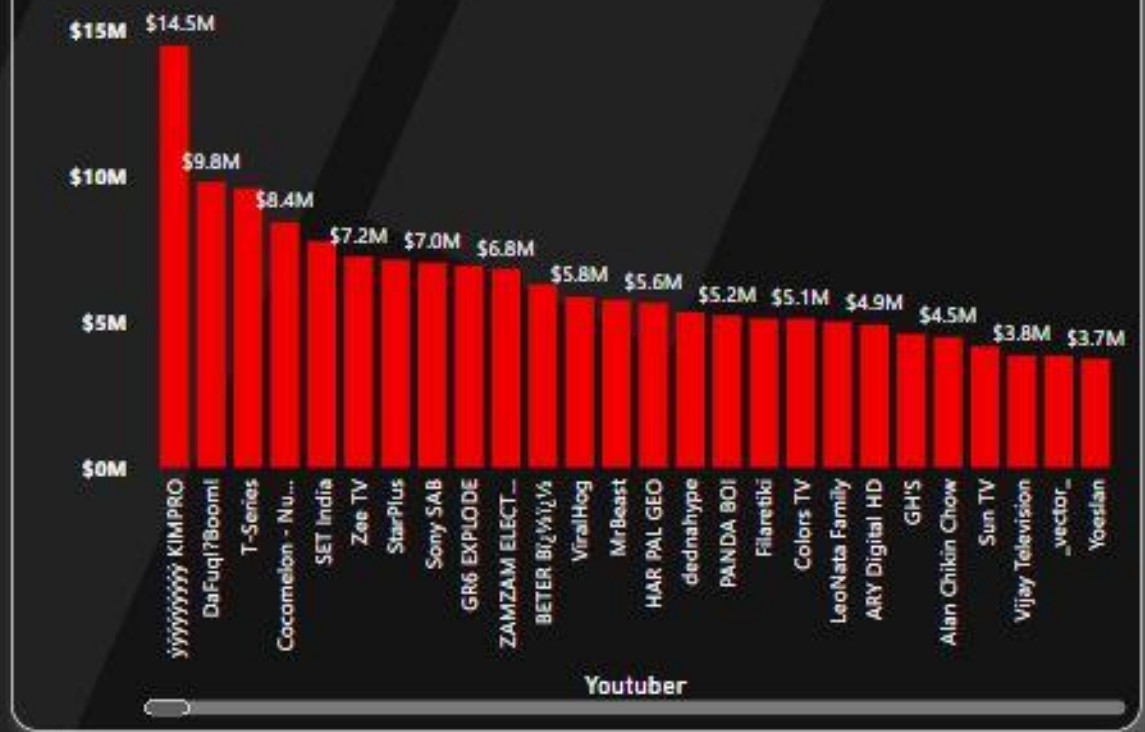
Total Yearly Earnings

$6.52bn

Total Monthly Earnings

$542.92M

Sum Monthly Earnings by YouTuber



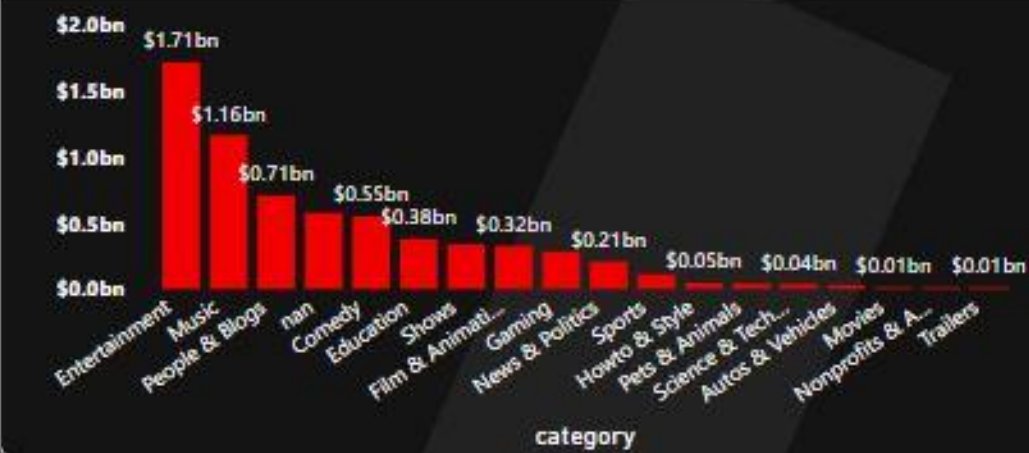
Top Yearly Earnings by Country



YouTuber

- BabyBus - C...
- BabyBus - K...
- Baim Paula
- BanderitaX
- BANGTANTV
- Bayashi TV
- BBC
- BBC News
- BBC News ...
- Beast Philan...
- Beast Reacts
- Bebefinn - ...

Sum Yearly Earnings by Category



Insight from the Earning Charts

- The total yearly earnings from the provided global YouTube statistics data amount to $6.52 billion.(card)
- Monthly earnings on YouTube based on the same data stand at $542.92 million.(card)
- The United States leads the yearly earnings chart for YouTubers with $2.1 billion, followed by India with $1.9 billion and South Korea with $0.3 billion. (Clustered Bar Chart)
- In terms of categories, Entertainment emerges as the highest-earning category with $1.71 billion, followed by Music with $1.16 billion, and People & Blog with $0.71 billion.(Clustered Column Chart)

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YouTube GEOGRAPHICAL ANALYTICS

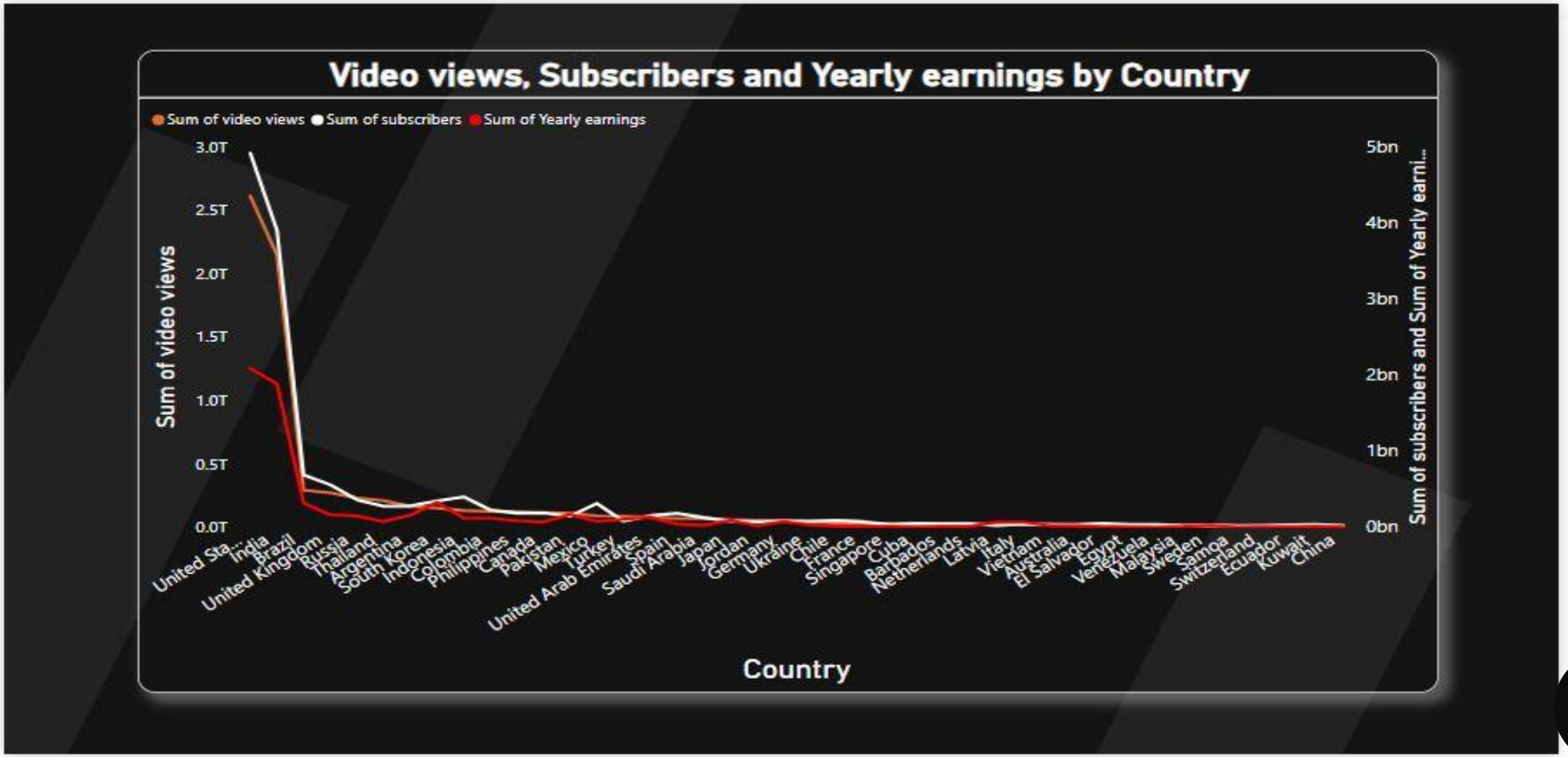
The YouTube GEOGRAPHICAL ANALYTICS visualization explores how geography impacts YouTube, addressing key questions such as total population based on data, yearly earnings by content category, monthly earnings for YouTubers, and identifying top earning countries. It's a tool to understand how where people are matters on YouTube, showing trends in earnings and population to paint a clear picture of YouTube's global reach and economic impact.

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Total Population
213bn
Total Urban Population
105bn
UnEmployment Rate %
8.93
Tertiary Education Enrollment
28.80K
Youtuber by Country
United States
India
Brazil
United King...
Indonesia
Mexico
Russia
Thailand
Argentina
Spain
Colombia
South Korea
Country
0
100
200
Count of Youtuber
Population by Country
1bn
0bn
Population
China
India
Unit...
Ind...
Paki...
Brazil
Rus...
Japan
Me...
Phili...
Egypt
Viet...
Tur...
Ger...
Thai...
Fra...
Unit...
Italy
Country
Total Uploads by Country
4M
2M
0M
TotalUploads
India
Unite...
Thaila...
Philip...
Indon...
Pakist...
Ukraine
Unite...
Brazil
Unite...
South...
Spain
Singa...
Turkey
Russia
Country
Video views, Subscribers and Yearly earnings by Country
Sum of video views
Sum of subscribers
Sum of Yearly earnings
Sum of video views
2T
0T
5bn
0bn
United ...
India
Brazil
United ...
Russia
Thailand
Argenti...
South K...
Indonesia
Colombia
Philippi...
Canada
Pakistan
Mexico
Turkey
United ...
Spain
Saudi A...
Japan
Jordan
Germany
Ukraine
Chile
France
Singap...
Country
Subscribers by Country
United States
India
Brazil
United...
Ind...
Ru...
South Ko...
Col...
Ca...
P...
S...
Mexico
Unit...
Argentina
Pakis...
Fr...
Thailand
Saud...

Insight

The line Chart shows a similarity in the video views, subscribers and Yearly Earnings in Countries, showing correlation among the three data column.



Other Insights

- Total population in consideration sum up to 213bn with close to half of them an urban population 1054bn
- 8.93% of the total population are unemployed according to the dataset.
- The United States is home to the majority of YouTubers with 197 of the Total 588 Youtubers.

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ANSWERS TO QUESTIONS CONSIDERED BEFORE THE ANLAYSIS AND VISUALIZATION

Here are the answers to specific questions considered before analysis and visualization:

1. Question: What insights can we gain about the activity and distribution of YouTubers on the platform?

- Answer: Our analysis reveals that there are 588 distinct YouTubers active on the YouTube App, with a majority of them located in the United States. This insight provides valuable information about the demographics and geographic spread of content creators on the platform.

2. Question: What trends in content categories resonate with audiences, and how does user interest impact content engagement and earnings?

- Answer: The entertainment industry emerges as the highest-earning category, followed by music, people, and blogs. Moreover, our analysis indicates that user interest in specific content categories, such as music and entertainment, significantly influences viewership and engagement levels. This suggests that understanding audience preferences is crucial for content creators to optimize their content strategy and maximize earnings.

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3. Question: Who are the influential YouTube creators from different countries, and how does their success contribute to the global YouTube landscape?

- Answer: Our data reveals that the United States leads the yearly earnings chart for YouTubers, followed by India and South Korea. Additionally, certain categories, such as entertainment, emerge as dominant in terms of earnings. This information allows us to identify influential YouTube creators from various countries and analyze their impact on a global scale, providing insights into regional differences and their contribution to the overall YouTube ecosystem.

4. Question: What factors influence a channel's earning potential, and how can we explore the correlation between channel performance metrics and estimated earnings?

- Answer: Subscriber count and content category selection play significant roles in determining a channel's earning potential. By analyzing metrics such as views, subscribers, and engagement rates in relation to estimated earnings, we can uncover patterns and correlations that provide insights into the factors influencing a channel's financial success.

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5. Question: How does the geographical location of YouTubers impact their earnings, and what insights can we derive from the distribution of successful channels across different regions?

- Answer: The analysis indicates that the geographical location of YouTubers influences their earnings, with the United States leading in yearly earnings followed by India and South Korea. Additionally, certain content categories may perform differently in different regions, impacting channel success and earnings. By visually representing the distribution of successful channels on a world map, we can uncover geographical trends and understand how location affects channel success and audience reach.

6. Question: What insights can we gain into content trends and their impact on viewer engagement and earnings?

- Answer: Certain categories, such as music and entertainment, demonstrate strong user preference and viewer engagement, while others like trailers and non-profit content are less profitable. By analyzing trends in content categories over time and correlating them with viewer engagement and earnings data, we can gain insights into how content trends evolve and their impact on audience engagement and channel earnings.

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LIMITATIONS

During my analysis of the global YouTube statistics data, I encountered several limitations that influenced the depth and scope of my insights. Here is a simplified breakdown:

1.Limited Context: I struggled with a lack of background information on how the data was collected, categorized, and the timeframe covered. This made it challenging to interpret findings accurately.

2. Data Completeness: There were gaps in the dataset, like missing variables or incomplete entries, which made it difficult to conduct thorough analysis and could lead to biased conclusions.

3. **Data Quality:** The data showed inconsistencies and inaccuracies, posing challenges to ensure the reliability and validity of my analysis.

4. Scope of Analysis: I found the analysis to be limited in scope, focusing on specific dimensions and overlooking broader industry trends, restricting the depth of understanding.

5. External Factors: It was hard to fully consider external factors like changes in YouTube algorithms, impacting the accuracy and relevance of my insights.

6. Geographical Visualization: Access to certain features like Map Display Visualization required a Power BI Services account, limiting the depth of geographic analysis.

- Acknowledging these limitations is crucial for understanding the reliability of the insights derived and ensuring transparency in the analysis process.

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RECOMMENDATION

Based on the analysis of global YouTube statistics data, several recommendations emerge to capitalize on opportunities and address challenges:

1.**Data Quality Improvement:** Enhance data quality assurance processes to address gaps in completeness and inconsistencies, ensuring more reliable and accurate insights for future analyses.

2.Content Strategy Optimization: Tailor content strategies based on the identified top-earning categories, such as entertainment and music, to align with viewer preferences and maximize engagement.

3.Geographic Expansion: Explore opportunities for geographic expansion, particularly in regions with significant YouTube user bases like the United States, India, and South Korea, to broaden audience reach and potential earnings.

4.Audience Engagement Initiatives: Implement initiatives to enhance audience engagement, such as leveraging subscriber counts to gauge category popularity and tailoring content to align with viewer interests.

5.Continuous Monitoring: Establish a system for continuous monitoring of industry trends, algorithm changes, and user behavior shifts to adapt content strategies and stay relevant in the dynamic digital landscape.



CONCLUSION

In conclusion, the analysis of global statistics data using various analytical techniques and tools has provided valuable insights into the dynamics of the YouTube platform. Through meticulous examination facilitated by DAX measures, we've gained a deeper understanding of trends, earnings patterns, and geographical influences. Despite encountering limitations, such as data completeness and platform constraints, the insights derived underscore the significance of data-driven decision-making in navigating the complexities of the digital landscape. Moving forward, leveraging these insights can inform strategic actions to optimize content strategies, enhance user engagement, and drive continued success in the dynamic world of online media.

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WANT TO SAY

THANK YOU